Intro to Statistics and Computation with Data

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Class 1, January 7, 2026

Marks breakdown for this course:

- Homework: 5
- Quiz: 15
- Midterm: 20
- Worksheet: 10
- Final: 50

Basic Syntax

```
#Creating a list of numbers
scores=c(21,23,34,54,12)
scores
```

```
## [1] 21 23 34 54 12
```

```
mean(scores)
```

```
## [1] 28.8
```

```
x=1:100
x[x>10|x<90]
```

```
##   [1]   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18
##  [19]  19  20  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36
##  [37]  37  38  39  40  41  42  43  44  45  46  47  48  49  50  51  52  53  54
##  [55]  55  56  57  58  59  60  61  62  63  64  65  66  67  68  69  70  71  72
##  [73]  73  74  75  76  77  78  79  80  81  82  83  84  85  86  87  88  89  90
##  [91]  91  92  93  94  95  96  97  98  99 100
```

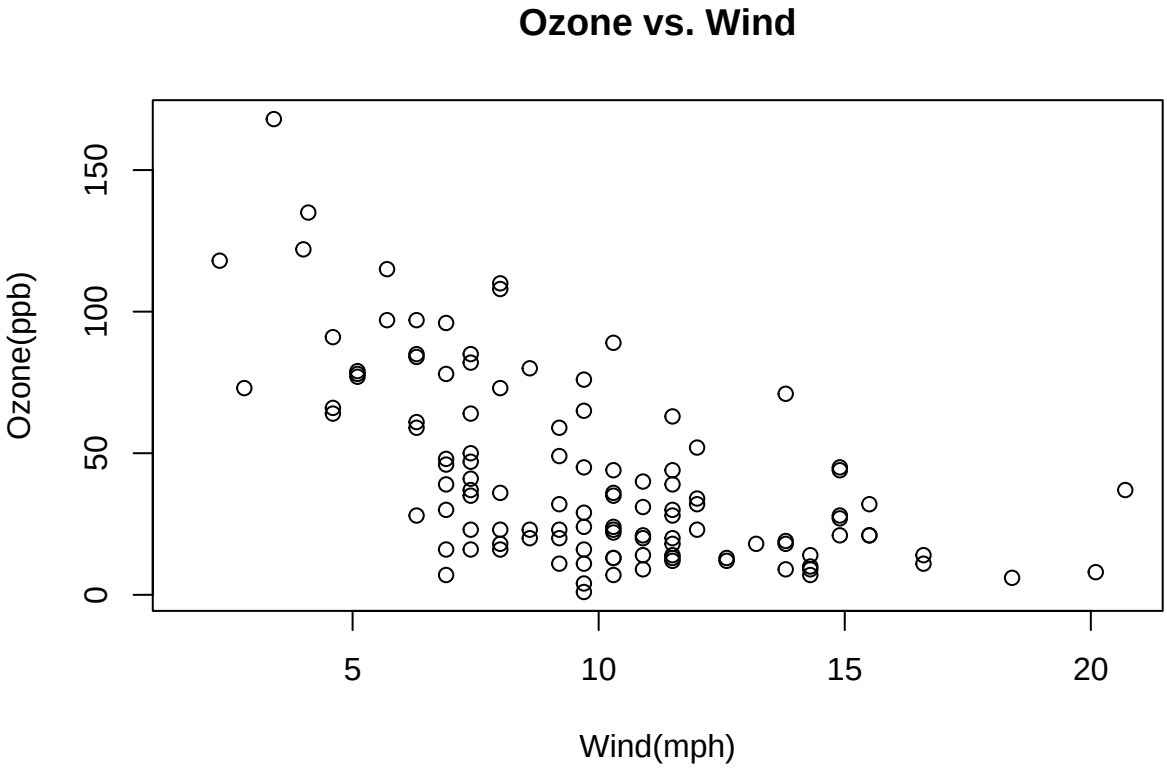
There are many pre-installed datasets in R.

```
#Air quality data of New York City
head(airquality,9)
```

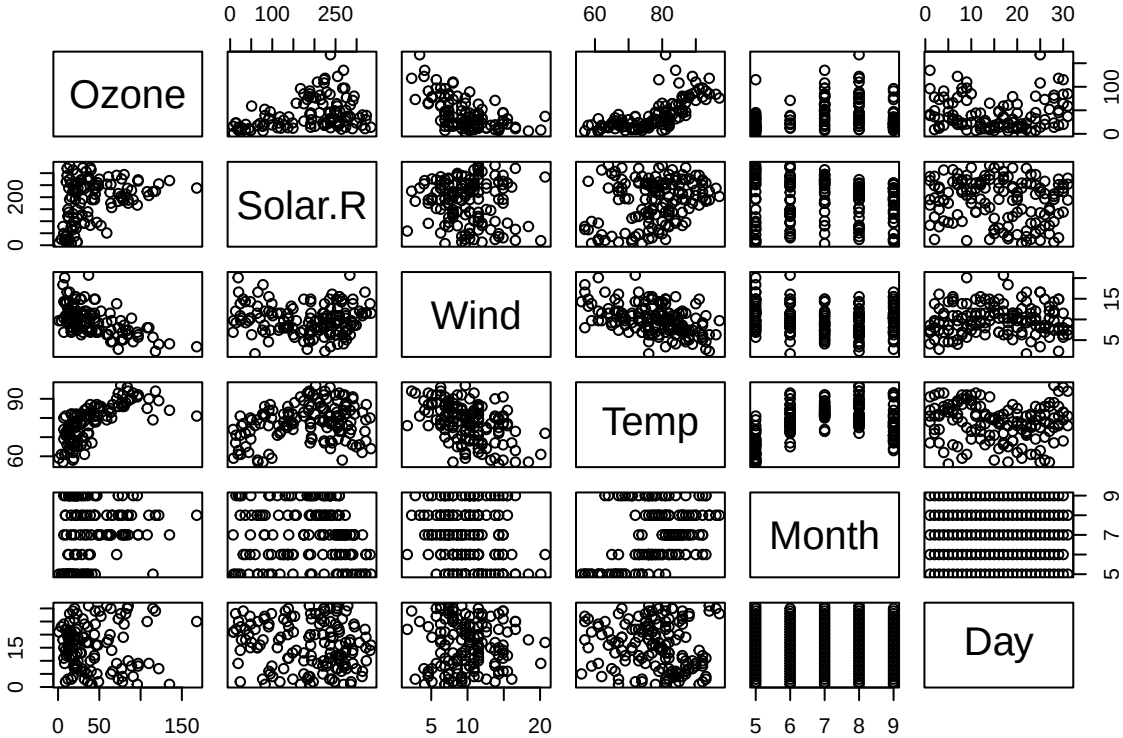
```
##   Ozone Solar.R Wind Temp Month Day
## 1    41     190  7.4   67     5   1
## 2    36     118  8.0   72     5   2
## 3    12     149 12.6   74     5   3
## 4    18     313 11.5   62     5   4
## 5    NA      NA 14.3   56     5   5
## 6    28      NA 14.9   66     5   6
## 7    23     299  8.6   65     5   7
## 8    19      99 13.8   59     5   8
## 9     8      19 20.1   61     5   9
```

Here's how we can plot graphs based on given data:

```
#Scatter plot of Ozone versus Wind
plot(airquality$Wind, airquality$Ozone,main="Ozone vs. Wind",xlab="Wind(mph)",ylab="Ozone(ppb)")
```

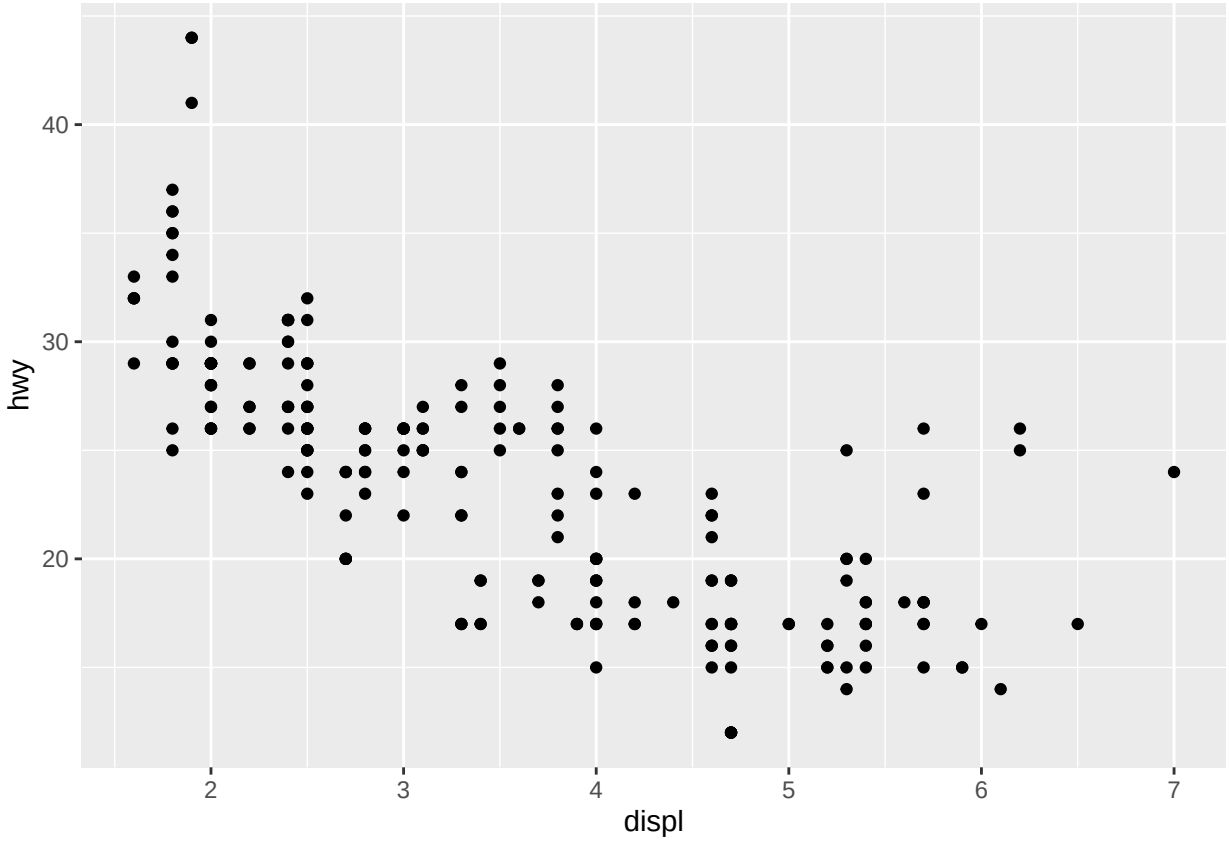



```
#What happens when you try to plot the whole dataset?
plot(airquality)
```

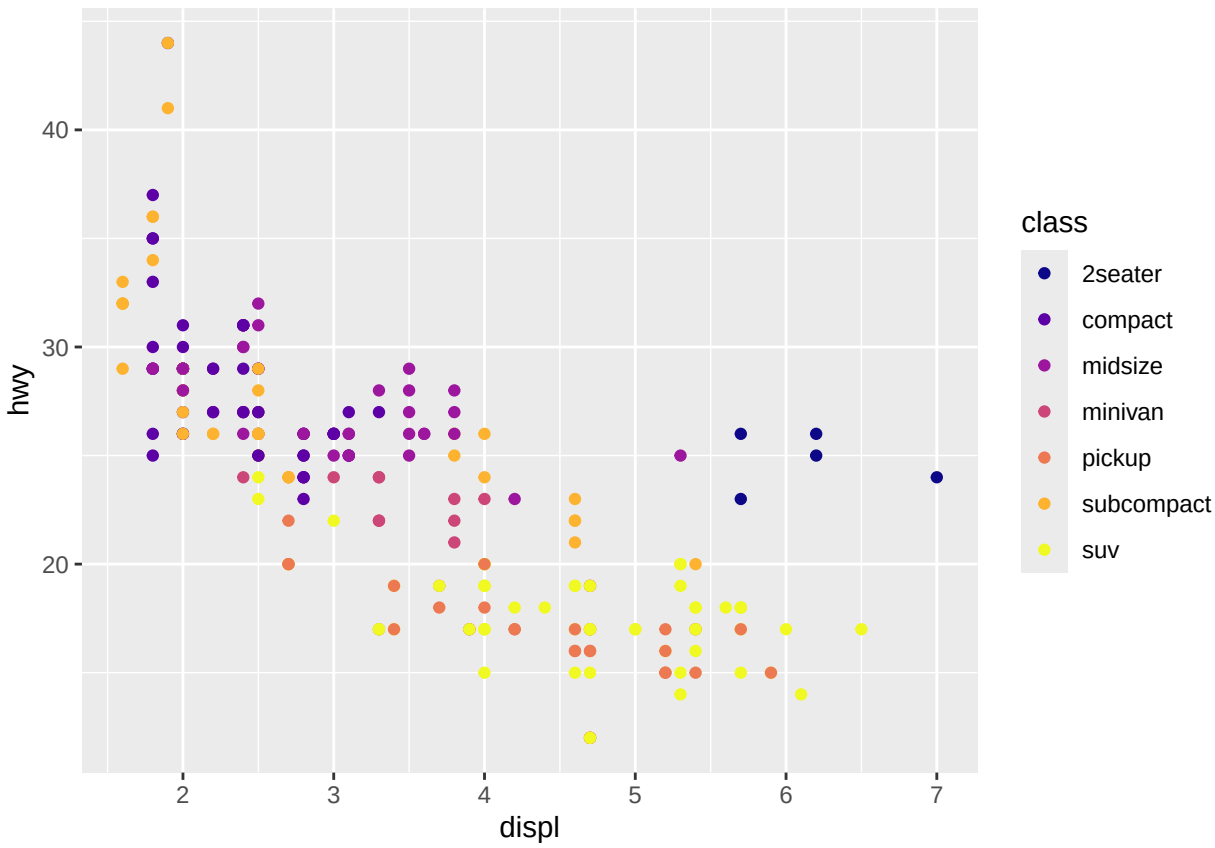


Class 2, January 14 2026

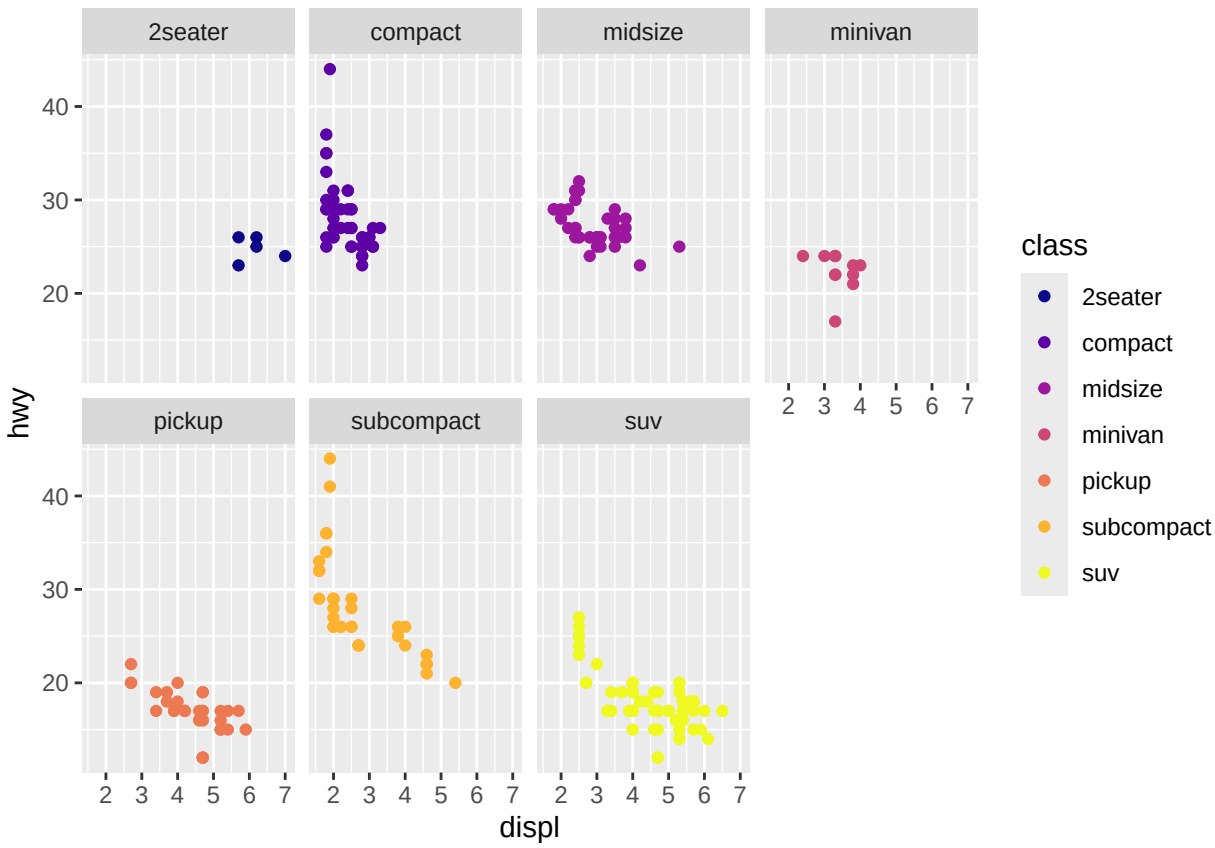
```
#Using ggplot
ggplot(data=mpg)+geom_point(mapping=aes(x=displ,y=hwy))
```



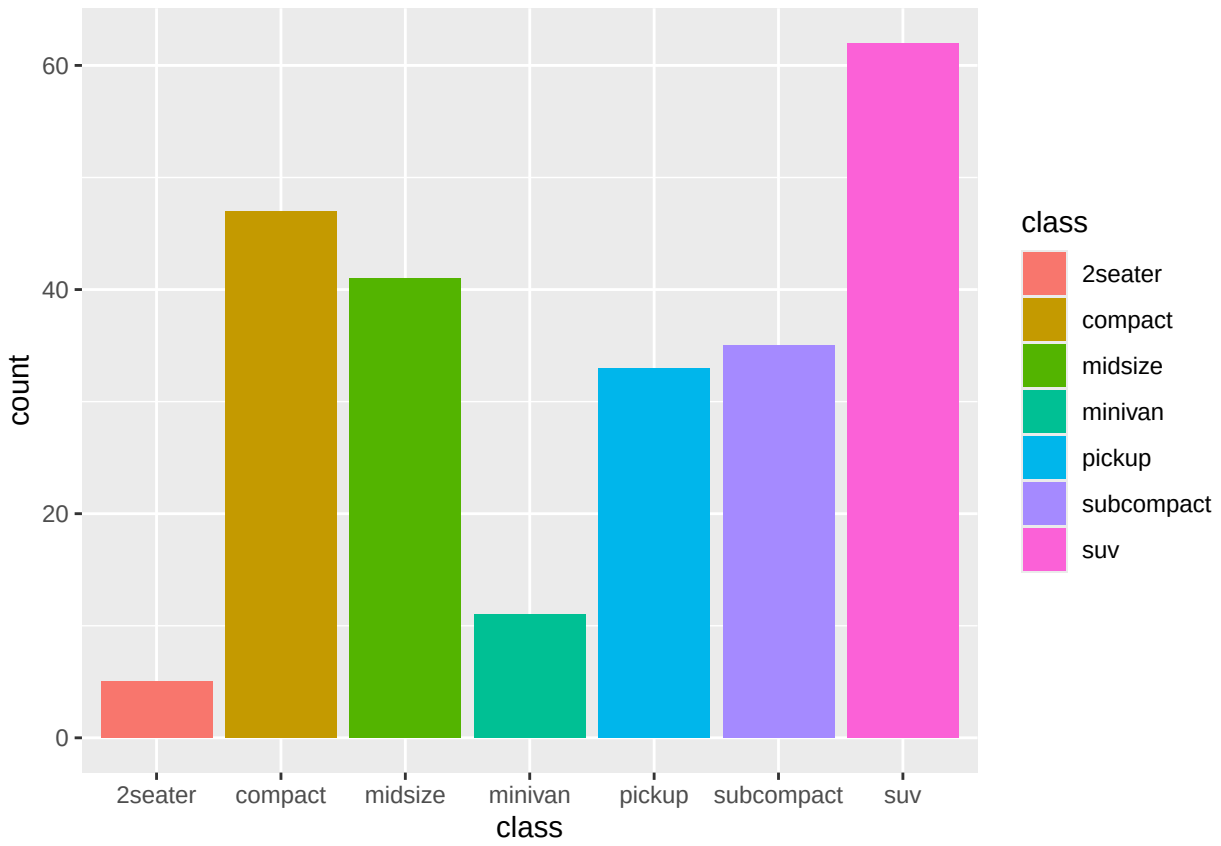

```
#Adding more layers
ggplot(data=mpg)+geom_point(mapping=aes(x=displ,y=hwy,colour=class))+scale_colour_viridis_d(option="pla
```



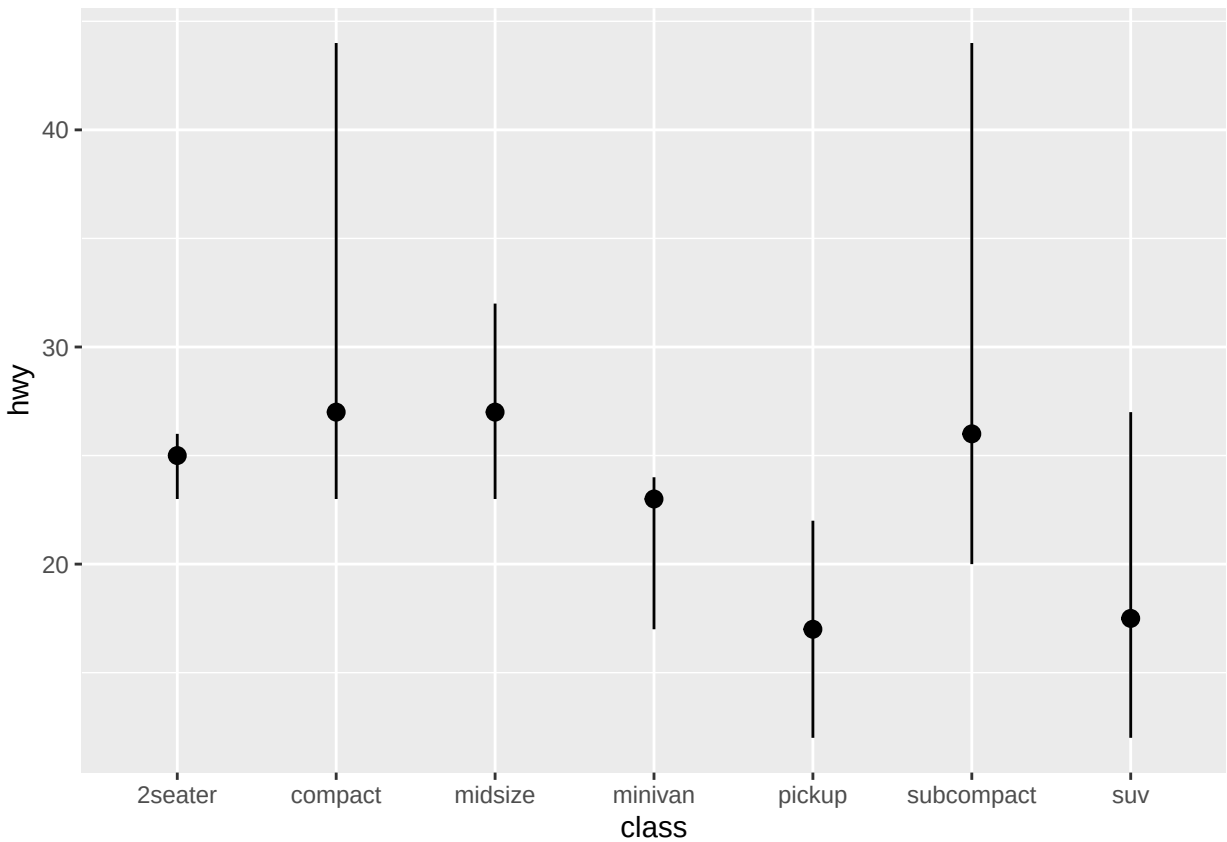

```
#Using facet wrap
ggplot(data=mpg)+geom_point(mapping=aes(x=displ,y=hwy,colour=class))+scale_colour_viridis_d(option="pla
```



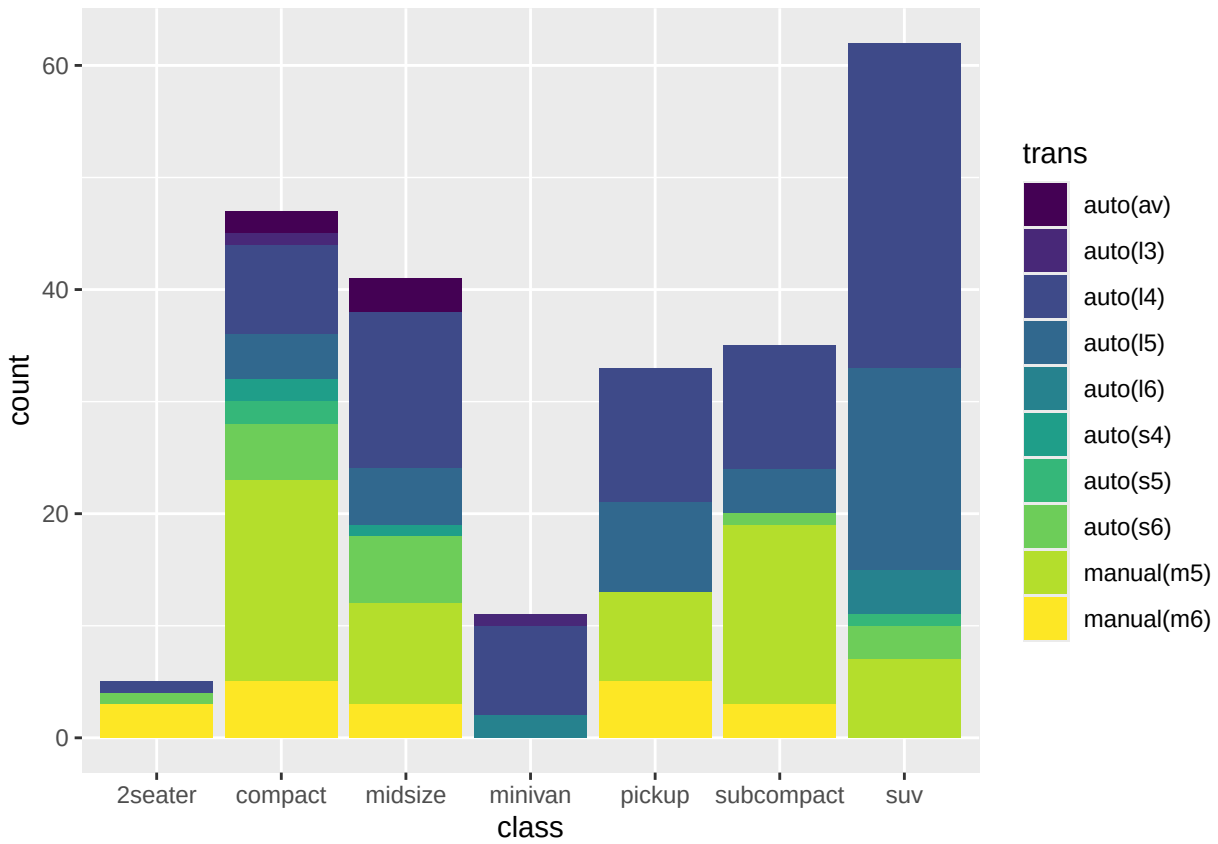

```r
#Using different plots
ggplot(data=mpg)+geom_bar(mapping=aes(x=class,fill=class))
```



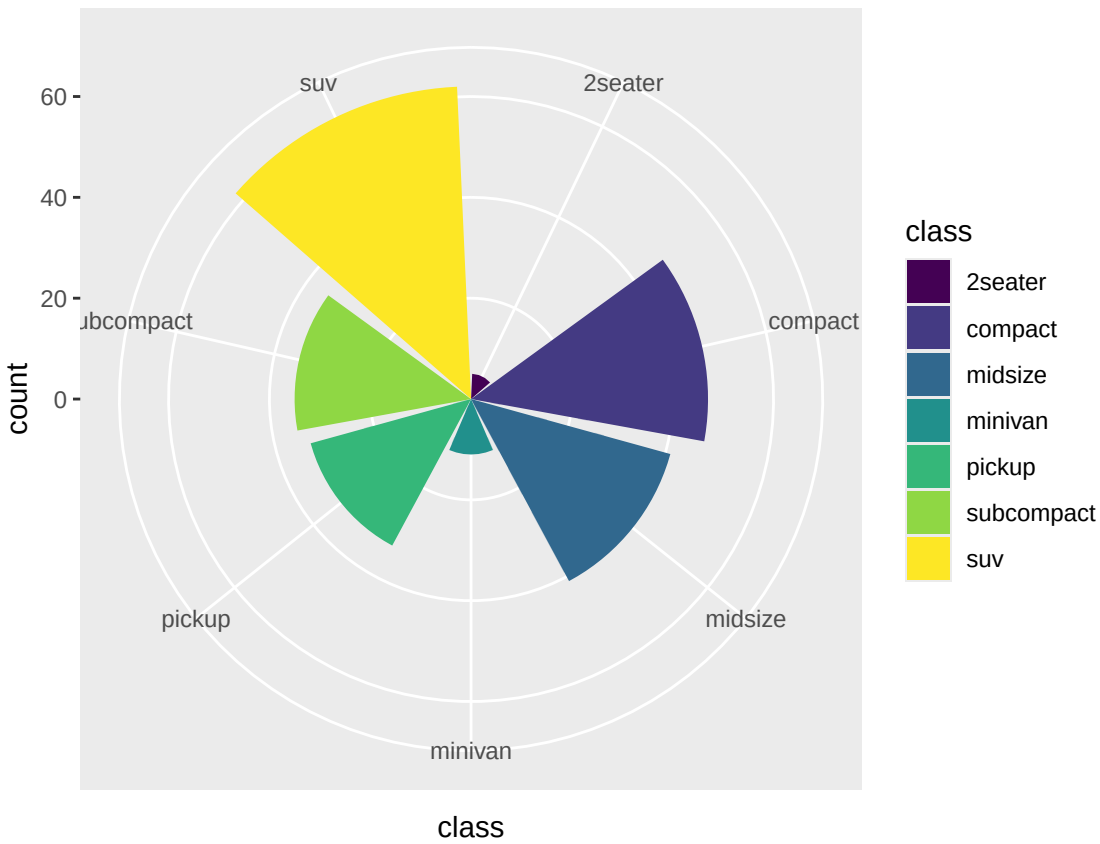

```r
#Summary
ggplot(data=mpg)+stat_summary(mapping=aes(x=class,y=hwy),fun.min=min,fun.max=max,fun=median)
```



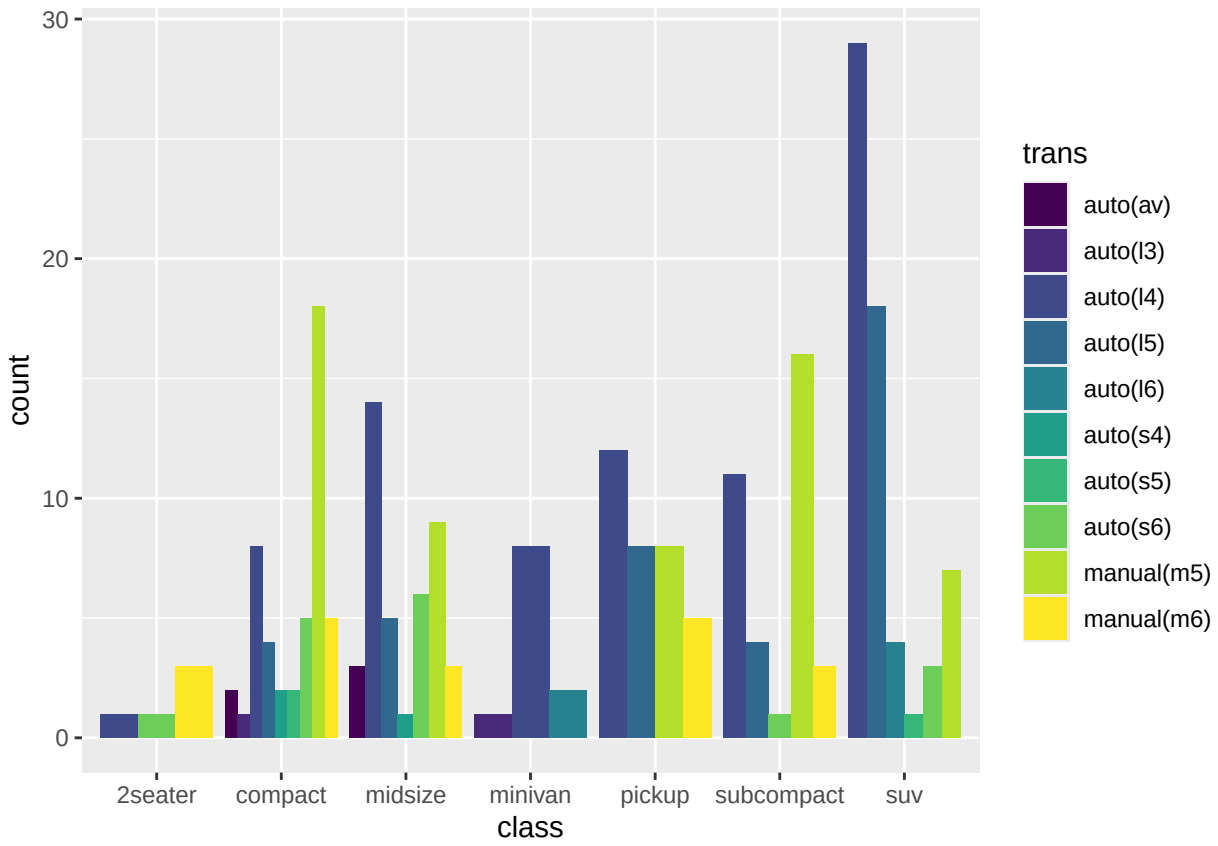

```r
#Adding more colour layers
ggplot(data=mpg)+geom_bar(mapping=aes(x=class,fill=trans))+scale_fill_viridis_d()
```



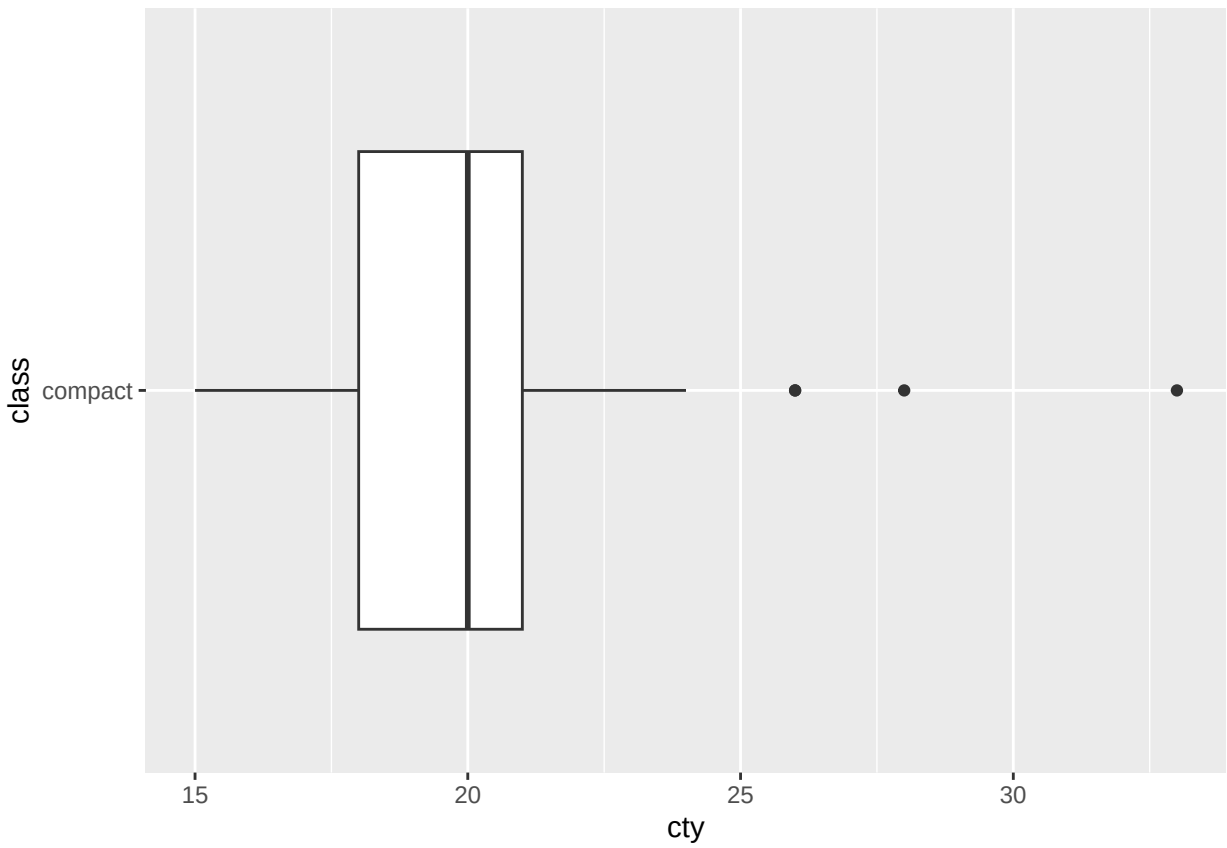

```r
#Making a pie chart
ggplot(data=mpg)+geom_bar(mapping=aes(x=class,fill=class))+scale_fill_viridis_d()+coord_polar()
```



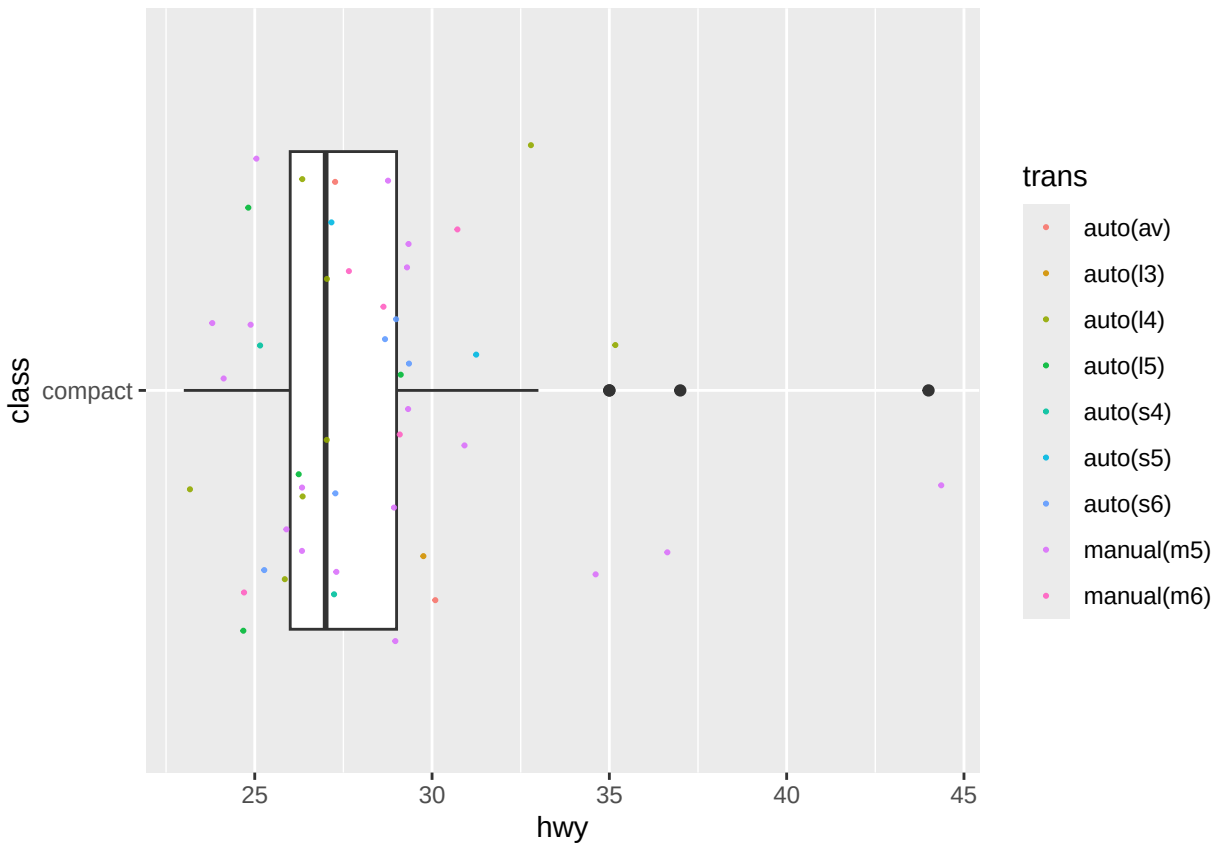

```
#Stacking bars
ggplot(data=mpg)+geom_bar(mapping=aes(x=class,fill=trans),position="dodge")+scale_fill_viridis_d()
```




```r
#BoxPlot- cty for compact
mpgcs=subset(mpg,mpg$class=="compact")
ggplot(data=mpgcs,mapping=aes(x=class,y=cty))+geom_boxplot()+coord_flip()
```




```
#Jitter
ggplot(data=mpgcs,mapping=aes(x=class,y=hwy))+geom_boxplot()+geom_jitter(aes(colour=trans),size=0.4,alpl
```



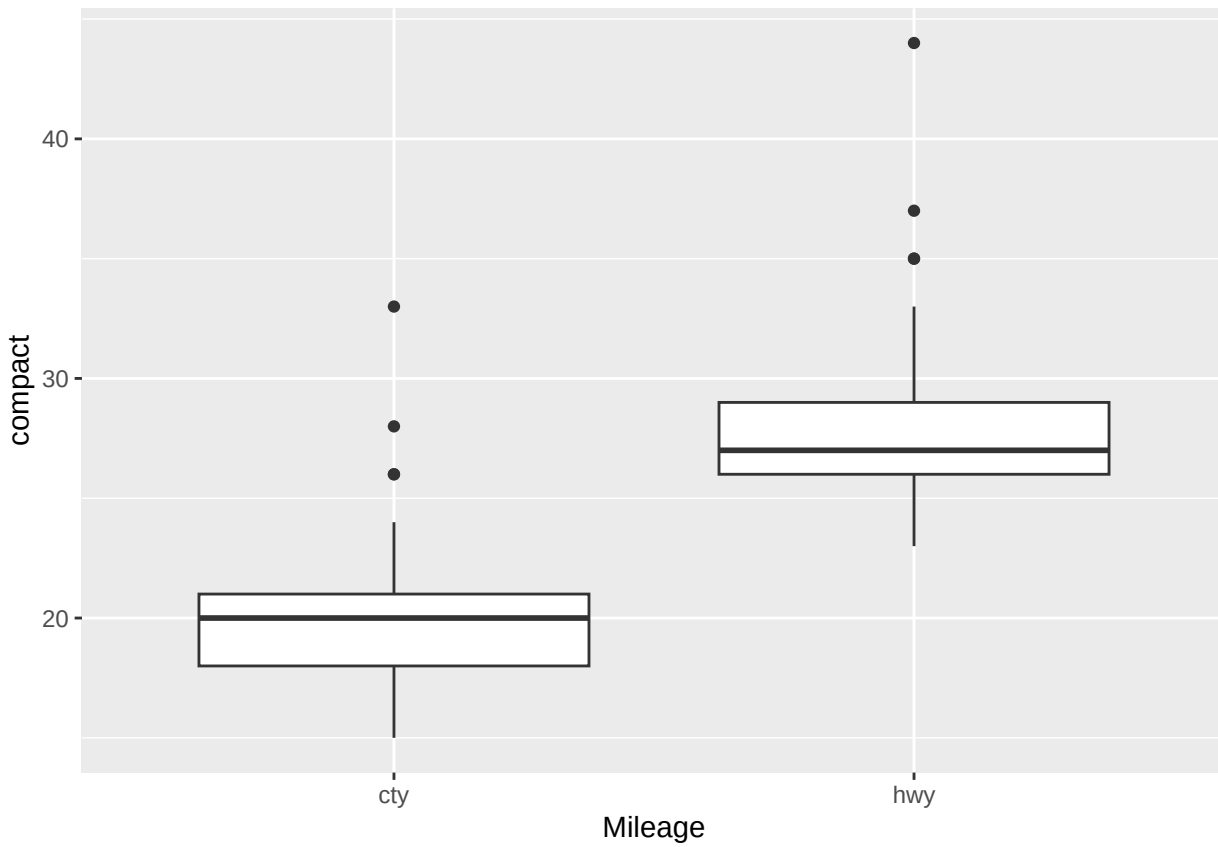

```
mpgr=mpg[c("class","hwy","cty")]
mpgrc=subset(mpgr,mpgr$class=="compact")
rdf<-data.frame(Mileage=c(rep("hwy",nrow(mpgrc)),rep("cty",nrow(mpgrc))),compact=c(mpgrc$hwy,mpgrc$cty))
head(rdf)
```

```
##   Mileage compact
## 1     hwy      29
## 2     hwy      29
## 3     hwy      31
## 4     hwy      30
## 5     hwy      26
## 6     hwy      26
```

```
tail(rdf)
```

```
##    Mileage compact
## 89     cty      22
## 90     cty      21
## 91     cty      21
## 92     cty      21
## 93     cty      16
## 94     cty      17
```

```
ggplot(data=rdf,mapping=aes(x=Mileage,y=compact))+geom_boxplot()
```



Now we see how to simulate various probability distributions in R

```
#Picking n many uniform random numbers
sample(1:6,2)
```

```
## [1] 6 3
```

```
#Rolling a dice ten times
sample(1:6,10,replace=T)
```

```
##  [1] 6 3 4 3 2 4 3 3 4 1
```

Class 3, January 28 2026

W would like to simulate various probability distributions in R. R can simulate both conitnuous and discrete uniform distributions as we saw in the previous class.

```
#Tossing a biased coin
sample(c(0,1),10,replace=T,prob=c(0.3,0.7))
```

```
##  [1] 0 1 0 1 0 1 1 1 1 1
```

```
#Simulating uniform random variables
set.seed(2021)
runif(3)
```

```
## [1] 0.4512674 0.7837798 0.7096822
```

```
#Simulating Bernoulli Random Variables
p=0.5
U=runif(1,min=0,max=1)
```

```r
X=as.numeric(U<p)
X
```

```
## [1] 1
```

```r
#Simulating Binomial Random Variables
#We want to similate a Bin(50,0.2) random variable.
p=0.2
U=runif(50,min=0,max=1)
X=as.numeric(sum(U)<10)
X
```

```
## [1] 0
```

```r
#Simulating a p1,p2,...,pn distribution
p1=0.35
p2=0.15
p3=0.4
p4=0.1
P=c(p1,p1+p2,p1+p2+p3,p1+p2+p3+p4)
X=c(1,2,3,4)
counter=1
set.seed(2021)
r=runif(1,min=0,max=1);
while(r>P[counter])
    counter=counter+1
end
```

```
## function (x, ...)
## UseMethod("end")
## <bytecode: 0x5630e88de178>
## <environment: namespace:stats>
```

```r
X[counter]
```

```
## [1] 2
```

```r
#Simulating Exponential Random Variables
U=runif(1,min=0,max=1)
lambda=1
X=-(1/lambda)*log(U)
X
```

```
## [1] 0.2436272
```

```r
#Simulating Gamma Random Variables
n=20;
lambda=10;
X=(-1/lambda)*sum(log(runif(n,min=0,max=1)))
X
```

```
## [1] 1.401866
```

```r
#Simulating Normal Random Variables (Box-Muller Transformations)
U1=runif(1)
U2=runif(1)
Z1=sqrt(-2*log(U1))*cos(2*pi*U2)
Z1=sqrt(-2*log(U1))*sin(2*pi*U2)
#Creating functions in R
```

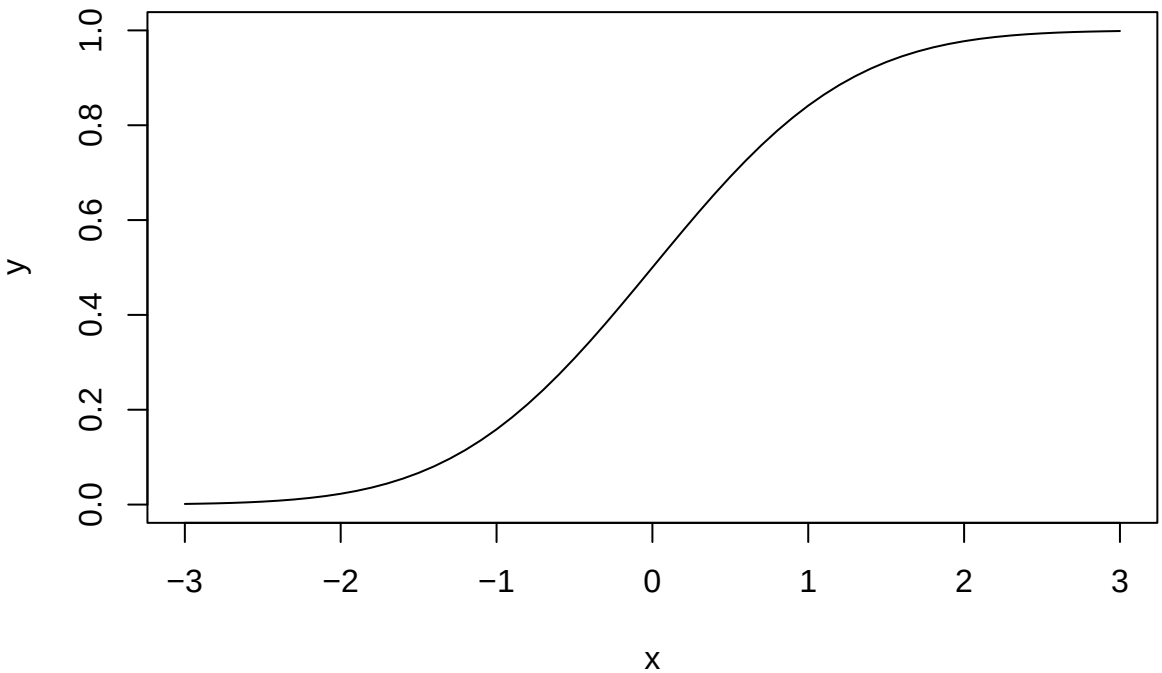
```r
f=function(x,y){
    x+y
}
f(2,2)
```

```
## [1] 4
```

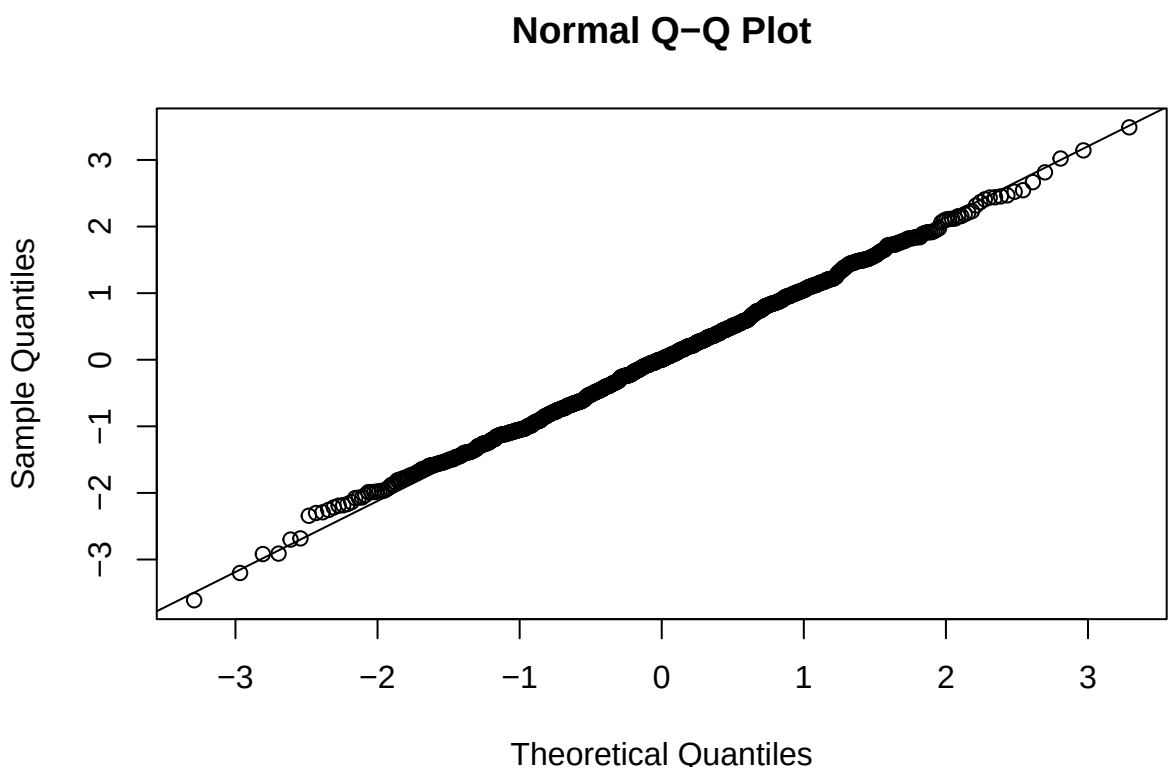
```r
f=function(x){dbinom(x,5,0.3)}
f(0:5)
```

```
## [1] 0.16807 0.36015 0.30870 0.13230 0.02835 0.00243
```

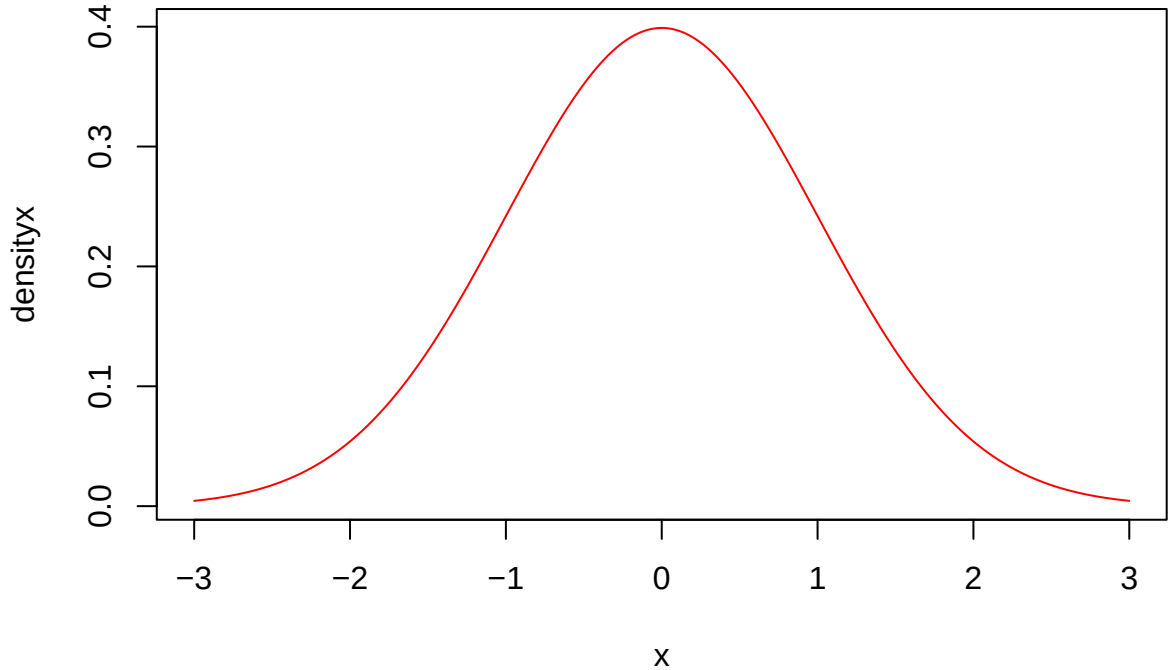
```r
#Plotting the CDF of Normal Distribution
x=seq(-3,3,by=0.1)
y=pnorm(x)
plot(x,y,type="l")
```




```r
#Getting the Q-Q PLot
x=rnorm(1000)
qqnorm(x)
qqline(x)
```

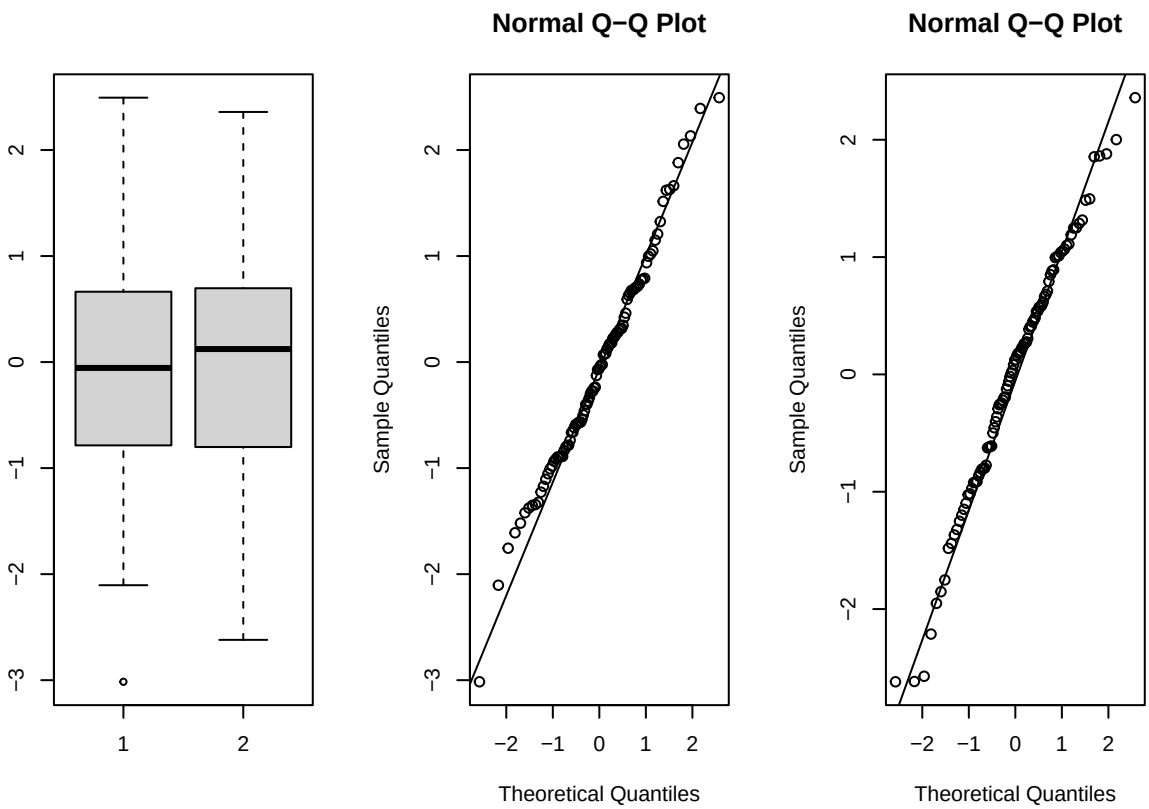



```
x=seq(-3,3,by=1/1000)
densityx=dnorm(x)
plot(x,densityx,type="l",col="red")
```

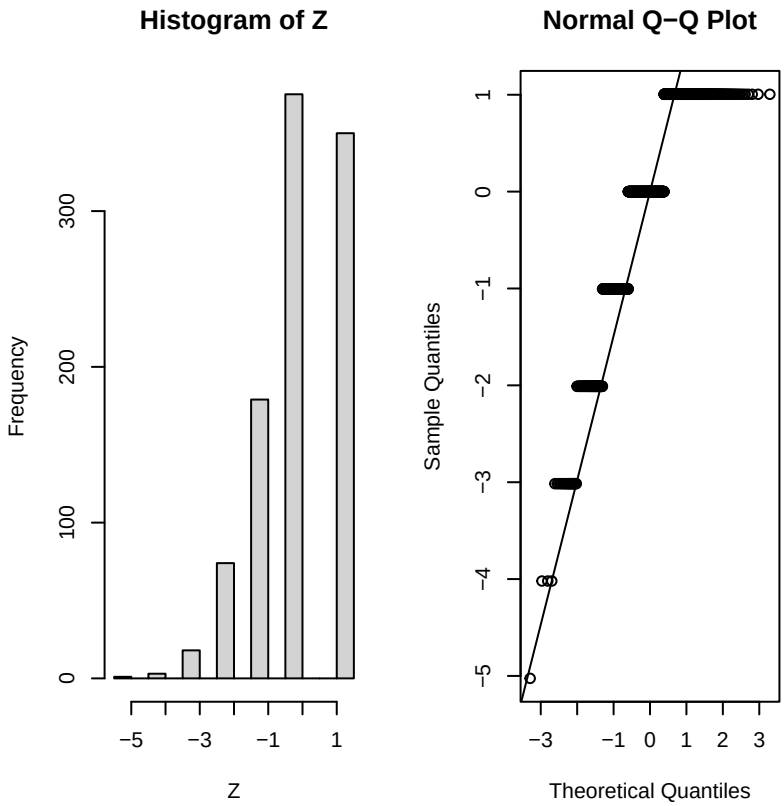



```
x=rnorm(100)
y=rt(100,30)
par(mfrow=c(1,3))
boxplot(x,y)
```

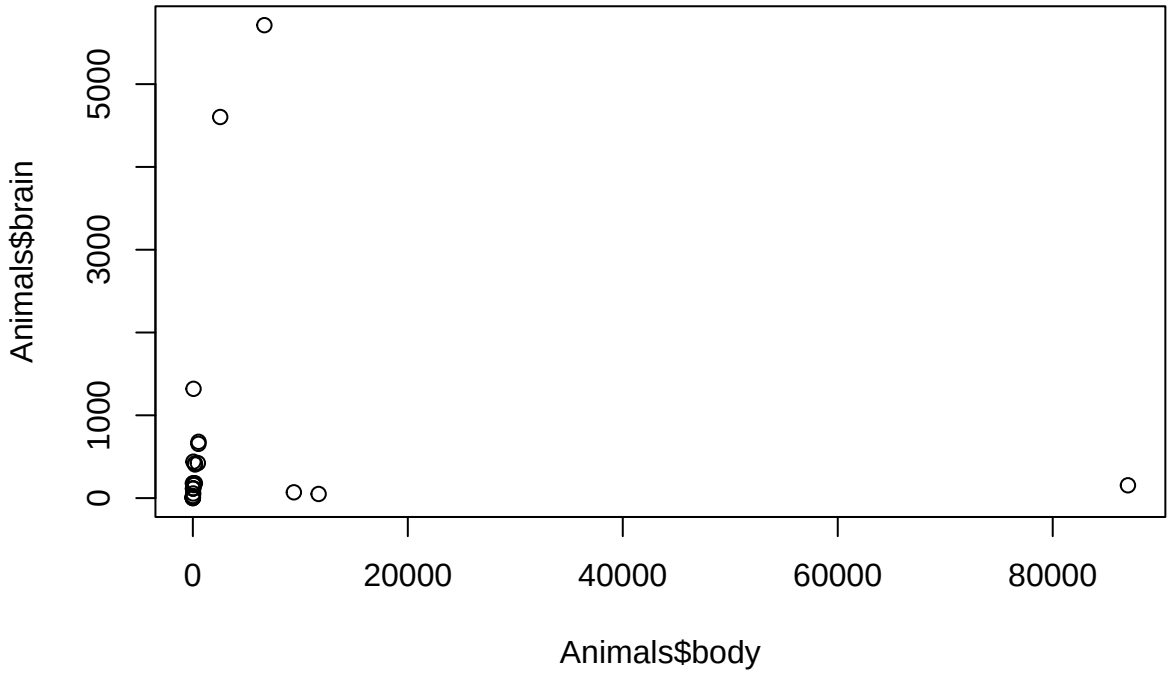
```r
qqnorm(x);qqline(x)
qqnorm(y);qqline(y)
```




```r
#Binomial CLT
N=1000
n=100
p=0.99
x=rbinom(N,n,p)
Z=(x-n*p)/sqrt(n*p*(1-p))
hist(Z)
qqnorm(Z)
qqline(Z)
#Using package MASS
par(mfrow=c(1,1))
```




```
plot(Animals$body,Animals$brain)
```




```
cor(rank(Animals$body),rank(Animals$brain))
```

```
## [1] 0.7162994
```